# Rooftop Segmentation from Aerial Imagery: Cross-Dataset Transfer and Applicability




## GROUP 10 Milestone 1

**Course:** DSAI Labs

**Team:** Team 10

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# 1. Problem Definition

**Problem statement:** Automatically detecting and segmenting building rooftops from high-resolution aerial (RGB) imagery. The task is to assign every pixel a label of rooftop or not rooftop, so the output is a mask from which rooftop extent and area can be derived. The aim is a model that works across different cities, building styles, lighting and image sources, not only on the data it was trained on.

**Primary stakeholders:**

- Urban planners and municipal bodies, who need up-to-date building maps and need to monitor unauthorized construction.
- Renewable energy planners and solar installers, who need to know where usable rooftop space exists.

**Secondary stakeholders:**

- Disaster management agencies (damage assessment, recovery planning).
- Drone navigation and logistics operators (identifying safe landing surfaces).
- Defense and surveillance users.
- Researchers who use building masks as input to downstream analysis.

**In scope:**

- Binary semantic segmentation of rooftops from RGB aerial or satellite-style imagery.
- Training and evaluation on public benchmark datasets.
- Comparison of performance across datasets with different resolution, geography and annotation style.
- A preliminary check on real-world imagery that was not part of the training data.

**Explicitly out of scope:**

- Individual building instance separation.
- Roof type, material or condition classification.
- 3D reconstruction or height estimation.
- Solar irradiance modelling, shading analysis, structural suitability or panel layout design.
- Real-time on-device deployment.

## 2. Why the Problem Matters and Where It Arises

Rooftop maps are a basic input for many decisions. Manual digitization is slow and expensive at city or national scale, and it goes out of date quickly.

- **Renewable energy:** Rooftop solar is a major route to distributed generation. Large-scale programmes such as India's PM Surya Ghar rooftop solar scheme need reliable estimates of where usable rooftops are (please verify current figures before quoting any numbers). Tools like Google's Project Sunroof show that automated rooftop analysis is a practical basis for solar planning.
- **Urban governance:** Comparing imagery over time helps detect unauthorized or unplanned construction.
- **Disaster response:** Fast building and roof maps support damage assessment and relief planning.
- **Drones and autonomous systems:** Aerial imagery can be used to find flat, safe surfaces for landing and delivery. The Inria benchmark paper itself mentions aerial imagery for autonomous driving and delivery as a motivating use case.

The problem arises wherever imagery is plentiful but labelled maps are not, which is common in rapidly growing cities and developing regions.

The task is hard for several reasons. Roofs vary greatly in shape, size, colour and material. Shadows, seasons and sensor differences change their appearance. Dense neighbourhoods have touching or occluded buildings, and trees, cars and rooftop objects confuse the boundaries. At oblique angles, relief displacement shifts roofs relative to the footprint on the ground.

## 3. How the Problem Is Currently Addressed

**3.1 Early approaches.** Early work used hand-crafted features (spectral, texture, edge and shape cues) with classical classifiers, plus morphological or object-based image analysis. These approaches break down on varied scenes because the features do not transfer between regions.

**3.2 Deep learning for segmentation.**

- Fully Convolutional Networks (Long et al., 2015) made end-to-end pixel-wise prediction practical.
- U-Net (Ronneberger et al., 2015) added an encoder-decoder with skip connections that preserve fine spatial detail. It became a standard baseline for remote sensing segmentation, and the Inria benchmark organizers' own follow-up work reported success with U-Net-style architectures and combined losses.
- DeepLabv3+ (Chen et al., 2018) added atrous (dilated) convolutions for multi-scale context.
- Instance-aware approaches such as Mask R-CNN (He et al., 2017) are used when individual buildings must be separated.

- Newer transformer and foundation-model approaches include SegFormer (Xie et al., 2021) and the Segment Anything Model (Kirillov et al., 2023). Their zero-shot behaviour on overhead imagery is still under study.

**3.3 Benchmarks and datasets.**

- **Massachusetts Buildings (Mnih, 2013):** An early, widely used building segmentation set covering the Boston area.
- **Inria Aerial Image Labeling (Maggiori et al., 2017):** It covers 810 km² at 0.3 m resolution with two classes, building and not building, spanning diverse urban environments. Its distinctive feature is that the test cities differ from the training cities, so it tests generalization rather than just within-image performance. purduearchives-ouvertes
- **AIRS (Chen et al., 2019):** A large-scale dataset at 7.5 cm resolution with over 220,000 buildings, where the task is defined as roof segmentation. The authors point out that most datasets provide building footprints as ground truth, which can be misaligned with roof outlines because of relief displacement. This matters for your problem, because footprint labels and true roof outlines are not the same thing. paperswithcodepaperswithcode
- **Other benchmarks:** SpaceNet (building footprint challenges), the WHU Building Dataset (Ji et al., 2019) and DeepGlobe (2018).
- **Large-scale building maps:** Microsoft Building Footprints and Google Open Buildings show that model-generated building maps can work at national scale. They are footprints rather than roof-specific masks.

**3.4 Common practice and gaps in how results are reported.** Many studies train and test within one dataset or even one scene. Maggiori et al. note that splitting a single image into training and test sets does not prove generalization to other inputs. Cross-dataset evaluation is therefore still less common than it should be. archives-ouvertes

## 4. Metrics and How Success Is Defined

- **Intersection over Union (IoU / Jaccard):** Overlap between predicted and true rooftop pixels divided by their union. This is the primary metric in building segmentation benchmarks, including Inria.
- **Precision, Recall and F1 (Dice):** Precision measures false alarms, recall measures missed rooftops, and F1 balances them. These are especially useful because rooftops are a minority class.
- **Pixel accuracy:** Reported often but weak on its own. Because most pixels are background, a model can score high accuracy while missing many buildings. It should only be read next to IoU and F1.
- **Object-level metrics:** Instance-level F1 or AP at an IoU threshold (such as AP@0.5), used when buildings are counted separately.
- **Boundary-aware metrics:** Boundary F-score or similar, used when edge precision matters.

In existing work, success usually means higher IoU on a held-out test set from a public benchmark. The stricter standard is performance on regions the model has never seen. Current top results on curated benchmarks are high, but they do not carry over reliably to new imagery.

## 5. Where Current Solutions Fall Short, and the Opportunity

1. **Generalization gap:** Models trained on one geography, resolution or sensor often degrade on another. Few studies compare datasets with very different characteristics under the same pipeline.
2. **Label inconsistency:** Datasets differ in ground truth definition (building footprint vs. roof outline), resolution and annotation quality. Reported scores are therefore not directly comparable across datasets, and the effect is rarely analysed.
3. **Class imbalance and weak minority-class performance:** Accuracy hides low recall on small or dense buildings.
4. **Gap between benchmarks and real use:** Benchmark data is curated. Operational imagery, such as government or municipal aerial data, differs in quality, resolution and labelling, and is rarely evaluated.
5. **Resource cost:** The strongest recent models are large, and there is limited guidance on what accuracy is achievable with a lighter, well-understood baseline.

**Opportunity:** A systematic, like-for-like evaluation of a standard segmentation approach across several public benchmarks, with a check on real-world imagery, gives a practical picture of how well rooftop detection transfers and where it fails. That evidence helps practitioners decide what is usable today.

## 6. Nature of the Contribution

The contribution lies mainly in robustness and generalization across datasets, and in a new deployment context (applying and testing on real-world, region-specific imagery). It is not a claim of a new state-of-the-art architecture. A secondary contribution is usability: producing masks from which rooftop area can be measured, which is the basis for downstream applications such as solar planning.

## 7. Evidence That Will Demonstrate the Approach Works

- **Quantitative:** IoU, precision, recall and F1 on held-out test sets for each benchmark, reported together and not through accuracy alone.

- **Generalization:** Performance under a consistent protocol across datasets that differ in resolution, geography and annotation style, including at least one test where the model is evaluated on data it was not trained on.
- **Baselines:** Comparison against published results on the same benchmarks and against simpler alternatives, to show the approach is competitive.
- **Qualitative:** Side-by-side visual comparisons of input, ground truth and prediction, including failure cases (shadows, dense blocks, unusual roofs).
- **Real-world check:** Inspection of predictions on imagery from outside the benchmarks, with an honest account of how performance changes.
- **Reproducibility:** Fixed data splits, documented preprocessing, and the same evaluation code for every dataset.

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